

Cell Specialisation

Test · 24 marks total

Name: _____

Date: _____

Score: _____ / 24

Section A — Specialised animal cells

A1. State one way a sperm cell is adapted for its function.

[2 marks]

A2. State one way a nerve cell is adapted for its function.

[2 marks]

A3. State one way a muscle cell is adapted for its function.

[2 marks]

Section B — Specialised plant cells

B1. State one way a root hair cell is adapted for its function.

[2 marks]

B2. State one way a xylem cell is adapted for its function.

[2 marks]

B3. State one way a phloem cell is adapted for its function.

[2 marks]

Section C — Explain in full

C1. Explain how the structure of a sperm cell is adapted to its function.

[3 marks]

C2. Explain how the structure of a xylem cell is adapted to its function.

[3 marks]

Section D — Unfamiliar cells

D1. A cell lining the lung has many cilia (tiny hair-like projections) on its surface that constantly beat back and forth. Suggest how this structure helps the cell's function.

[3 marks]

D2. A gland cell that secretes hormones into the blood contains an unusually large number of mitochondria. Suggest why.

[3 marks]